



Sydney Girls High School

MATHEMATICS EXTENSION 1

YEAR 11 ASSESSMENT TASK 3 2019

Time Allowed: 60 minutes + 5 minutes reading time

Total: 60 marks

Topics: Permutations and combinations; The binomial expansion and Pascal's triangle; Inequalities; Graphical relationships; Polynomials; Inverse functions; Parametric form of a function or relation; Further trigonometric identities.

General Instructions:

- There are FOUR (4) Questions which are of equal value.
- Attempt all questions.
- Show all necessary working. Marks may be deducted for badly arranged work or incomplete working.
- A correct answer without any working will be awarded a maximum of 1 mark.
- Start each question on a new page.
- Write on one side of the paper only.
- Diagrams are NOT to scale.
- Board-approved calculators may be used.
- Write using a black or blue pen.
- White-out or correction tape is NOT to be used.

Student Name : _____

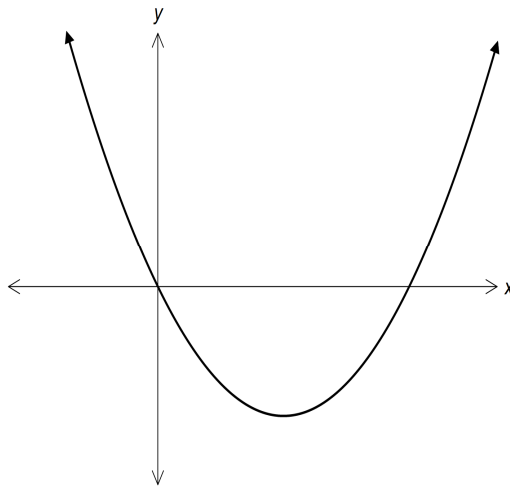
Teacher Name : _____

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- (a) How many different arrangements of the letters of the word MATHS are possible? 1
- (b) Simplify $\frac{\sin 2x}{1 + \cos 2x}$. 2
- (c) A curve has parametric equations $x = 2t$ and $y = 3 - t^2$. 2
Find the Cartesian equation of the curve.
- (d) The polynomial $P(x)$ is given by $P(x) = 2x^3 + 7x^2 - 46x + 21$.
- (i) Show that $P(x)$ is divisible by $(x - 3)$. 1
- (ii) Hence, fully factorise $P(x)$. 3
- (e)
- (i) State the domain and range of $y = 3 \sin^{-1}(2x)$. 2
- (ii) Hence, sketch $y = 3 \sin^{-1}(2x)$. 1
- (f) Solve the inequality $\frac{2x}{x-1} \geq 1$. 3

End of Question 1

(a)



The graph of the parabola $y = f(x)$ is given above. Sketch on separate number planes the graphs of:

(i) $y = |f(x)|$. 1

(ii) $y = \frac{1}{f(x)}$. 2

(b) Show that $\cos 3\theta = 4\cos^3 \theta - 3\cos \theta$. 2

(c) A debating group consists of 12 students, 8 of whom are girls.
How many debating teams of three students can be formed if:

(i) There are no girls at all. 1

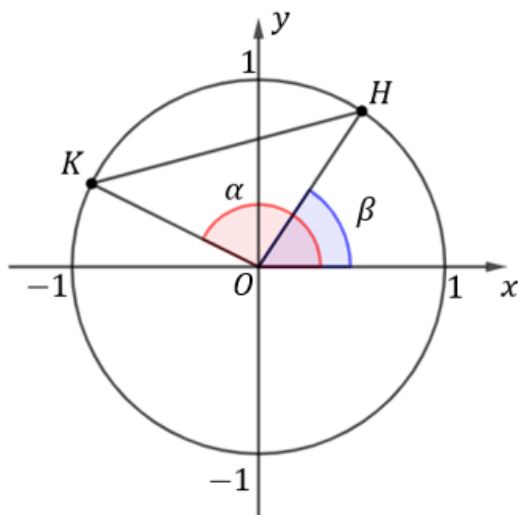
(ii) There is exactly one girl. 1

(iii) There is at least one girl. 2

Question 2 continues on the next page.

- (d) The diagram below shows the circle $x^2 + y^2 = 1$, where the point H has coordinates $(\cos \beta, \sin \beta)$ and the point K has coordinates $(\cos \alpha, \sin \alpha)$.

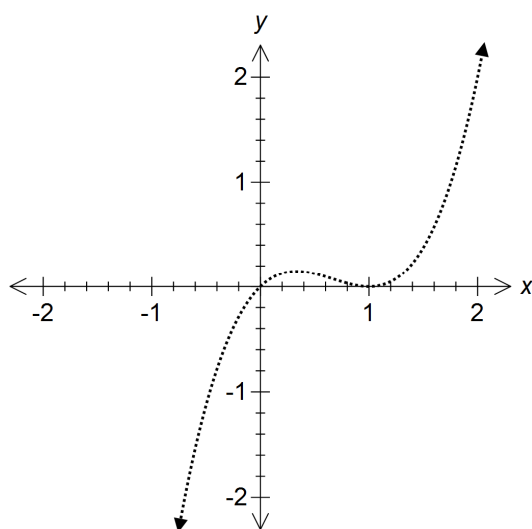
The line OH makes an angle of β with the positive x -axis and the line OK makes an angle of α with the positive x -axis.



- | | | |
|-------|---|---|
| (i) | Find an expression for the distance HK . | 1 |
| (ii) | By considering the triangle KOH , show that $HK^2 = 2 - 2\cos(\alpha - \beta)$. | 1 |
| (iii) | Hence show that $\cos(\alpha - \beta) = \cos \alpha \cos \beta + \sin \alpha \sin \beta$. | 2 |
| (iv) | If $\alpha = \frac{\pi}{3}$ and $\beta = \frac{\pi}{4}$ find the exact value of $\cos \frac{\pi}{12}$. | 2 |

End of Question 2

- (a) Find the term independent of x in the expansion $\left(x^2 + \frac{2}{x}\right)^{15}$. 2
- (b) Find an expression for y in terms of x if for $x > 0$ and $y > 0$, $\tan^{-1} x = \tan^{-1} y + \frac{\pi}{4}$. 3
- (c)
- (i) Given the function $f(x) = 6x - x^2$, find the largest possible domain including zero, such that an inverse function $f^{-1}(x)$ exists. 1
- (ii) Find the equation of $f^{-1}(x)$. 3
- (d) Simplify $\frac{\cos(\theta + \alpha) - \cos(\theta - \alpha)}{\sin(\theta + \alpha) + \sin(\theta - \alpha)}$. 2
- (e) In the binomial expansion of $\left(1 + \frac{x}{k}\right)^n$, the coefficient of x^3 is twice the coefficient of x^2 . Prove that $n = 6k + 2$. 2
- (f) The graph of the $y = f(x)$ is given below. Sketch the graph of $y^2 = f(x)$. 2



End of Question 3

- (a) The polynomial equation $x^3 + bx^2 + cx + d = 0$ has roots α , α^2 and α^3 for some real number $\alpha \neq 0$.

(i) Find in terms of b , c and d the value of $\frac{1}{\alpha} + \frac{1}{\alpha^2} + \frac{1}{\alpha^3}$. 2

(ii) Show that $b^3d - c^3 = 0$. 2

- (b) By substituting a suitable value for x in the expansion $(1+x)^n$, show that 2

$$1 + 2\binom{n}{1} + 4\binom{n}{2} + \dots + 2^{n-1}\binom{n}{n-1} = 3^n - 2^n.$$

- (c) Show that $\frac{\sin \theta + \sin(90^\circ - \theta) + 1}{\sin \theta - \sin(90^\circ - \theta) + 1} = \cot \frac{\theta}{2}$. 3

- (d) A monic cubic polynomial when divided by $x^2 + 7$ leaves a remainder of $x + 12$ and when divided by x leaves a remainder of -6 . 3

Find the polynomial in the form $ax^3 + bx^2 + cx + d$.

- (e) (i) State the domain of $y = x^2 \cos^{-1} x$. 1

- (ii) Hence, sketch the graph of $y = x^2 \cos^{-1} x$. 2

End of paper

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2019
Yr 11 Ext. 1 Task # 3

Question 1

(a) $5! = 120$ /1

(b) $\frac{\sin 2x}{1 + \cos 2x}$

$= \frac{2 \sin x \cos x}{1 + 2 \cos^2 x - 1}$

$= \frac{\sin x}{\cos x}$

$= \tan x$ /2

(c) $x = 2t$ ①

$y = 3 - t^2$ ②

From ①: $t = \frac{x}{2}$ ③

Sub ③ in ②:

$y = 3 - \left(\frac{x}{2}\right)^2$

$y = 3 - \frac{x^2}{4}$

$y = -\frac{x^2}{4} + 3$ /2

(d)(i)

$P(x) = 2x^3 + 7x^2 - 46x + 21$

$P(3) = 2(27) + 7(9) - 46(3) + 21$
 $= 0$

$\therefore (x-3)$ is a factor

$\therefore P(x)$ is divisible by $(x-3)$

d) ii) $\begin{array}{r} 2x^2 + 13x - 7 \\ x-3 \overline{) 2x^3 + 7x^2 - 46x + 21} \\ \underline{2x^3 - 6x^2} \end{array}$

$13x^2 - 46x$

$\underline{13x^2 - 39x}$

$-7x + 21$

$\underline{-7x + 21}$

0

$\therefore P(x) = (x-3)(2x^2 + 13x - 7)$

$= (x-3)(x+7)(2x-1)$
/3

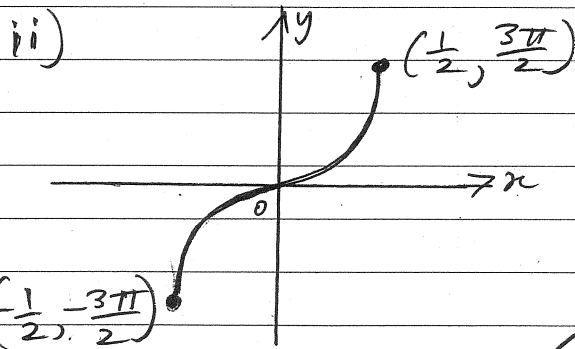
e) i) $y = 3 \sin^{-1} 2x$

D: $-1 \leq 2x \leq 1$

$-\frac{1}{2} \leq x \leq \frac{1}{2}$

R: $-\frac{\pi}{2} \leq \frac{y}{3} \leq \frac{\pi}{2}$

$-\frac{3\pi}{2} \leq y \leq \frac{3\pi}{2}$ /2



(f) $\frac{2x}{x-1} \times (x-1)^2 \geq 1 \times (x-1)^2$

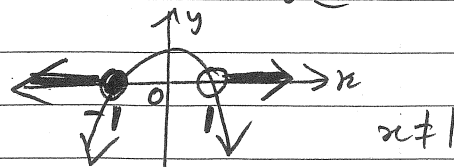
$2x(x-1) \geq (x-1)^2$

$(x-1)^2 - 2x(x-1) \leq 0$

$(x-1)[(x-1) - 2x] \leq 0$

$(x-1)(-x-1) \leq 0$

$-(x-1)(x+1) \leq 0$



$x \leq -1$ OR $x \geq 1$ /3

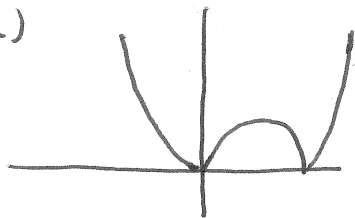
Notes for Q1

e) ii) Shape of curve needs to be more defined
(more "curvy")

f) - Test for Quadratic Inequality
needs to be more clearly shown

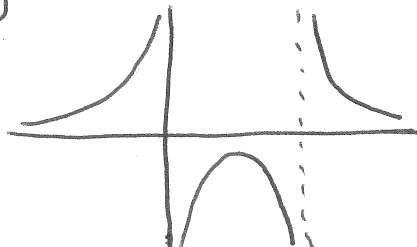
- Many students forgot the restriction $x \neq 1$

2 cos (i)



This graph must have sharp corners on the x axis

(ii)



(h) $\cos 3\theta = \cos (2\theta + \theta)$

$$\begin{aligned} &= \cos 2\theta \cos \theta - \sin 2\theta \sin \theta \\ &= (\cos^2 \theta - \sin^2 \theta) \cos \theta - 2 \sin \theta \cos \theta \sin \theta \\ &= \cos^3 \theta - \sin^2 \theta \cos \theta - 2 \sin^2 \theta \cos \theta \\ &= \cos^3 \theta - 3 \sin^2 \theta \cos \theta \\ &= \cos^3 \theta - 3(1 - \cos^2 \theta) \cos \theta \\ &= \cos^3 \theta - 3 \cos \theta + 3 \cos^3 \theta \\ &= 4 \cos^3 \theta - 3 \cos \theta \quad \text{G.E.D.} \end{aligned}$$

In show questions important steps must not be left out.

(c) (i) $+ C_3 = 4$

(ii) $8x + C_2 = 48$

(iii) $12C_3 - 4 = 216$

(d) (i) $\sqrt{(\cos \alpha - \cos \beta)^2 + (\sin \alpha - \sin \beta)^2}$

Co-ordinate geometry is required here.

(ii) $Hk^2 = 1^2 + 1^2 - 2 \times 1 \times 1 \times \cos (\alpha - \beta)$
 $= 2 - 2 \cos (\alpha - \beta)$

The cosine rule is required here.

(iii) $2 - 2 \cos (\alpha - \beta) = \left(\sqrt{(\cos \alpha - \cos \beta)^2 + (\sin \alpha - \sin \beta)^2} \right)^2$
 $= \cos^2 \alpha - 2 \cos \alpha \cos \beta + \cos^2 \beta + \sin^2 \alpha - 2 \sin \alpha \sin \beta + \sin^2 \beta$
 $= 2 - 2 \cos \alpha \cos \beta - 2 \sin \alpha \sin \beta$

$\therefore \cos (\alpha - \beta) = \cos \alpha \cos \beta + \sin \alpha \sin \beta$

G.E.D.

$$(iv) \cos \frac{\pi}{12} = \cos \left(\frac{\pi}{3} - \frac{\pi}{4} \right)$$

$$= \cos \frac{\pi}{3} \cos \frac{\pi}{4} + \sin \frac{\pi}{3} \sin \frac{\pi}{4}$$

$$= \frac{1}{2} \times \frac{1}{\sqrt{2}} + \frac{\sqrt{3}}{2} \times \frac{1}{\sqrt{2}}$$

$$= \frac{1 + \sqrt{3}}{2\sqrt{2}}$$

Q3 Ext 1 2019

a) $T_{k+1} = {}^{15}C_k (x^2)^{15-k} \cdot 2^k x^{-k}$

$$x^{30-2k-k} = x^0$$

$$x^{30-3k} = x^0$$

$$30-3k=0$$

$$k=10$$

$$T_{11} = {}^{15}C_{10} (x^2)^5 \cdot 2^{10} \cdot x^{-10}$$

$$= 3075072$$

This question
was done well.

b) $\tan^{-1}x - \tan^{-1}y = \frac{\pi}{4}$

Let $\alpha = \tan^{-1}x$

$\tan \alpha = x$

$B = \tan^{-1}y$

$\tan B = y$

$$\tan(\alpha - B) = \tan \frac{\pi}{4}$$

$$\frac{\tan \alpha - \tan B}{1 + \tan \alpha \tan B} = 1$$

$$\frac{x - y}{1 + xy} = 1$$

$$\frac{x - y}{1 + xy} = 1$$

$$x - y = 1 + xy$$

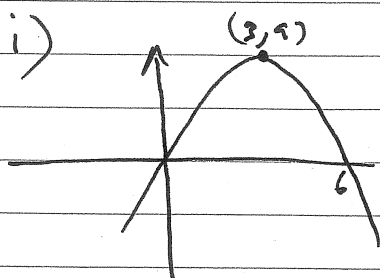
$$x - 1 = xy + y$$

$$x - 1 = y(x + 1)$$

$$y = \frac{x - 1}{x + 1}$$

This question
was done
poorly so many
students didn't
know where to
start.

c) i)



$$D: x \leq 3$$

* Many students didn't know the correct domain

ii) $x = 6y - y^2$

$$y^2 - 6y + 9 = 9 - x$$

$$(y - 3)^2 = 9 - x$$

* Many students didn't choose the correct branch

$$y = 3 \pm \sqrt{9 - x}$$

$$x \leq 3 \quad \therefore y = 3 - \sqrt{9 - x}$$

d)
$$\frac{-2 \cancel{\sin \theta} \sin \alpha}{\cancel{2} \sin \theta \cos \alpha}$$

$$= -\tan \alpha$$

Some students used the wrong signs $\therefore$ didn't get the correct answer

e)
$$2 {}^nC_2 \frac{n^2}{k^2} = {}^nC_3 \frac{n^3}{k^3}$$

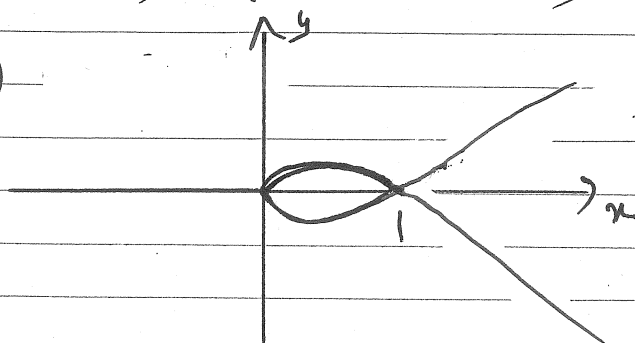
$$\frac{2n!}{(n-2)! 2! k^2} = \frac{n!}{(n-3)! 3! k^3}$$

* This question was done poorly

$$6k^3 (n-3)! = (n-2)! k^2$$

$$6k(n-3)! = (n-2)(n-3)! \quad \therefore n = 6k + 2$$

f)



* Many students couldn't draw the correct graph

Q4

$$a) x^3 + bx^2 + cx + d = 0$$

$$\alpha + \alpha^2 + \alpha^3 = -b$$

$$\alpha(\alpha^2) + \alpha(\alpha^3) + \alpha^2\alpha^3 = c$$

$$\alpha(\alpha^2)(\alpha^3) = -d$$

$$i) \frac{1}{\alpha} + \frac{1}{\alpha^2} + \frac{1}{\alpha^3}$$

$$= \frac{\alpha^2\alpha^3 + \alpha\alpha^3 + \alpha\alpha^2}{\alpha\alpha^2\alpha^3} = -\frac{c}{d}$$

$$ii) b^3d - c^3 = 0$$

$$b^3d = c^3$$

$$[-(\alpha + \alpha^2 + \alpha^3)]^3 [-\alpha^6] = [\alpha^2(\alpha + \alpha^2 + \alpha^3)]^3$$

$$\alpha^6(\alpha + \alpha^2 + \alpha^3)^3 = \alpha^6(\alpha + \alpha^2 + \alpha^3)^3$$

This is true

$$\therefore b^3d = c^3$$

$$\text{OR } b^3d - c^3 = 0$$

$$b) (1+x)^n = {}^nC_0 + {}^nC_1x + {}^nC_2x^2 + \dots + {}^nC_{n-1}x^{n-1} + {}^nC_nx^n$$

Subs $x=2$

$$3^n = {}^nC_0 + 2{}^nC_1 + 4{}^nC_2 + \dots + 2^{n-1}{}^nC_{n-1} + 2^n$$

$$3^n - 2^n = {}^nC_0 - 2{}^nC_1 + 4{}^nC_2 + \dots + 2^{n-1}{}^nC_{n-1}$$

proven.

Q4

$$c) \frac{\sin \theta + \sin(90 - \theta) + 1}{\sin \theta - \sin(90 - \theta) + 1} = \cot \frac{\theta}{2}$$

$$\text{LHS} = \frac{\sin \theta + \cos \theta + 1}{\sin \theta - \cos \theta + 1}$$

$$= \frac{\frac{2t}{1+t^2} + \frac{1-t^2}{1+t^2} + 1}{\frac{2t}{1+t^2} - \left(\frac{1-t^2}{1+t^2}\right) + 1}$$

$$= \frac{2t + 1 - t^2 + 1 + t^2}{2t - 1 + t^2 + 1 + t^2}$$

$$= \frac{2(1+t)}{2t(1+t)} = \frac{1}{t} = \frac{1}{\tan \frac{\theta}{2}}$$

$$= \cot \frac{\theta}{2} = \text{RHS}$$

Q4

d) $p(x) = (x^2 + 7)(ax + b) + x + 12$

$$p(x) = ax^3 + bx^2 + 7ax + 7b + x + 12$$

$$p(x) = ax^3 + bx^2 + (7a + 1)x + 7b + 12$$

$$p(x) = x^3 + bx^2 + 8x + 7b + 12 \quad (\text{Monic poly})$$

$$p(0) = -6$$

$$7b + 12 = -6$$

$$b = -\frac{18}{7}$$

$$\therefore p(x) = x^3 - \frac{18}{7}x^2 + 8x - 6$$

e) $y = x^2 \cdot \cos^{-1} x$

$$D: -1 \leq x \leq 1$$

$$R: 0 \leq y \leq \pi$$

